Paleomagnetic reexamination of the Lower Ordovician Wabana and Bell Island Groups of the Avalon Peninsula of Newfoundland

K. L. BUCHAN
Geomagnetic Research Laboratory, Department of Physics, Memorial University of Newfoundland, St. John's, Nfld., Canada A1B 3X7
AND
J. P. HODYCH
Department of Physics and Department of Geology, Memorial University of Newfoundland, St. John's, Nfld., Canada A1B 3X7

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The Wabana and Bell Island Groups of the Avalon Peninsula of Newfoundland contain oolitic hematite beds of Early Ordovician age, dipping ~11° north-northwest. Twenty-one oolitic hematite samples from nine sites were thermally demagnetized in 10 or 11 steps to 660 or 680°C. At 450°C, 15 samples from six sites define a stable magnetization direction ($D = 21.2°$, $I = -12.5°$, $k = 137$, $\alpha_{95} = 5.7°$) with respect to bedding. The remaining six samples failed to attain the stable end-point, five of them because of growth of much "magnetite," which probably resulted from breakdown of siderite during thermal demagnetization.

The paleopole for the Wabana – Bell Island Groups lies at 33°N 102°E (dm = 5.8°, dp = 3.0°). It is compared with other early Paleozoic paleopoles from cratonic North America and the Avalon zone, and its significance for the Iapetus Ocean is discussed.

Les groupes de Wabana et de Bell Island de la péninsule d'Avalon de Terre-Neuve renferment des couches d'hématite oolithique du Ordovicien inférieur, pendant ~11° vers le nord-ouest. Vingt-et-un échantillons d'hématite oolithique de neuf sites ont été désaimantés par traitement thermique en 10 ou 11 étapes jusqu'à 660 ou 680°C. À 450°C, 15 échantillons correspondant à six sites ont fourni une direction stable de désaimantation ($D = 21{,}2°$, $I = -12{,}5°$, $k = 137$, $\alpha_{95} = 5{,}7°$) par rapport à la stratification. Les six autres échantillons n'ont pu procurer une valeur stable: la raison pour cinq d'entre eux est la croissance de beaucoup de "magnétite" résultant probablement de la dissociation de la sidérite au cours du traitement thermique de désaimantation.

Le paléopôle des groupes Wabana – Bell Island se situe à 33°N 102°E (dm = 5,8°, dp = 3,0°). Il est comparé avec d'autres paléopôles du Paléozoïque inférieur des cratons de l'Amérique du Nord et de la zone d'Avalon et on y discute son importance par rapport à l'Océan Iapetus.

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Introduction

The Wabana and Bell Island Groups of the Avalon Peninsula of Newfoundland consist of fossiliferous Lower Ordovician shales and sandstones with many intercalated oolitic hematite beds, all dipping gently to the northwest. The oolitic hematites were first studied paleomagnetically by Rao (1970), who reported a well defined remanence direction after thermal demagnetization to 685°C. The corresponding paleopole at 28°N, 192°E (Deutsch and Rao 1970, 1977)—significantly to the east of other Ordovician paleopoles for North America—has been used to position the Avalonian microcontinent in early Paleozoic continental reconstructions (e.g., Scotese *et al.* 1979). The only other early Paleozoic paleopole then available for the Avalon zone was from the red beds of the Ratcliffe Brook Formation of New Brunswick (Black 1964), but this pole was not considered very reliable because the red beds had only been subjected to alternating field (AF) demagnetization.

Rao (1970) also reported paleopoles from rather scattered normal and reversed natural remanent magnetization (NRM) directions (46°N, 89°E and 51°N, 121°E, respectively). These were thought to represent late Paleozoic remagnetizations. These poles, however, have not been referred to in subsequent publications (Deutsch and Rao 1970, 1977).

A recent paleomagnetic study by Hodych and Pätzold (1980) of the folded Point Lance sills of the Avalon Peninsula yielded a probable Late Cambrian paleopole not vastly different from those for North America. This led us to reexamine the paleomagnetic data reported for the Wabana Group. We suspected that the ores might be anisotropic (though Rao (1970) thought this unlikely)

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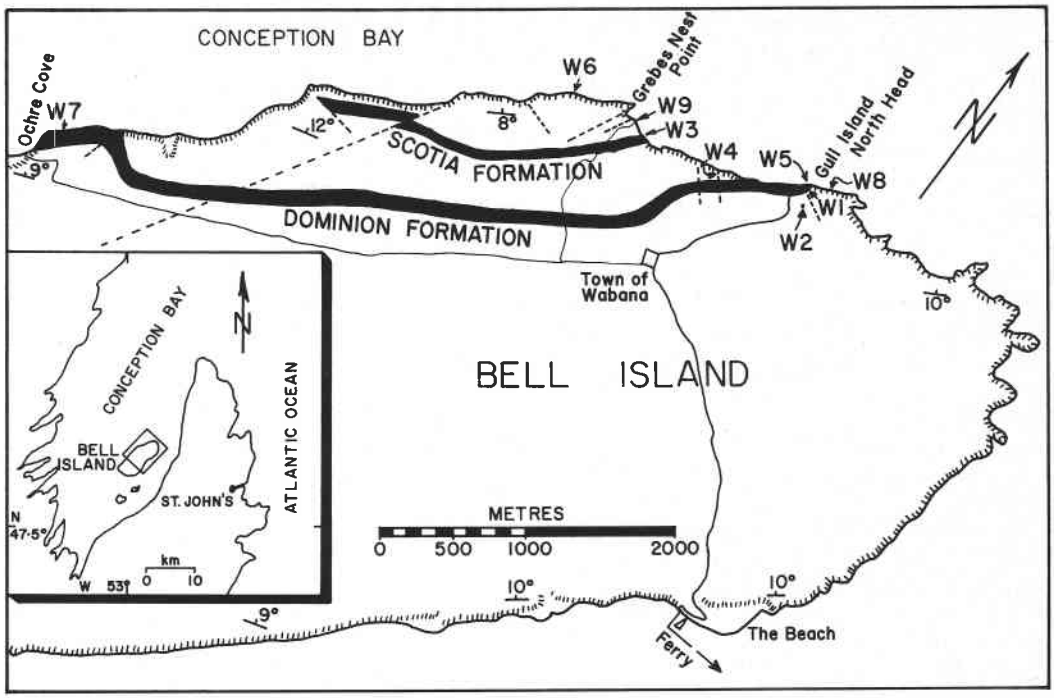


FIG. 1. The sampling sites on Bell Island, Avalon Peninsula, Newfoundland. Geology after Ranger (1979).

with the plane of bedding easier to magnetize, because in Rao's (1970) thermal demagnetizations the remanences typically began with a northerly or southerly direction close to the bedding plane, swung counterclockwise close to the bedding plane, and ended at a peak temperature of 685°C with an easterly direction close to the bedding plane. This hypothesis was supported by one of the two other published paleomagnetic studies of oolitic hematites: the Silurian Clinton Formation of Alabama had been found to have a higher magnetic susceptibility in the bedding plane than at right angles to it, and its remanence lay close to the bedding plane (Howell *et al.* 1958, 1960).

Oriented samples of Wabana Group and Bell Island Group oolitic hematites were collected and thermally demagnetized prior to testing for anisotropy of acquisition of thermal remanence. However, we could not reproduce the directional results of the earlier study. Instead, the natural remanent magnetization (NRM) initially directed toward the north-northeast with a shallow inclination was found to move systematically upwards about 12° out of the bedding plane upon thermal demagnetization. (No examples of the shallow southerly NRM reported by Rao (1970) were observed, though of his five sites with reverse NRM, three sites—his only sites from the Dominion Formation—were in the general vicinity of our sites W1, W5, and W8, and one site—his only site from the Scotia Formation—was in the general vicinity of our site W3.) In many samples, thermal demagnetization became difficult at temperatures above 600°C because the growth of "magnetite" during heating introduced a large spurious component. Nevertheless, most of the samples yielded a well defined paleopole not very different from the somewhat scattered Ordovician paleopole for cratonic North America. We found no evidence of anisotropy of acquisition of thermal remanence.

Geology and sampling sites

Twenty-one oriented blocks were collected from oolitic hematite beds at the nine sites shown in Fig. 1 (geology after Ranger 1979) and listed in Table 1. Sites W1, W5, W7, and W8 are in the Dominion Formation, an ore bed approximately 4 m thick at the very top of the Bell Island Group. Site W2 is in a thin bed a few metres below the Dominion Formation. The following sites are in the overlying Wabana Group: site W4 is in a thin bed not far above the Dominion Formation; site W3 is in the Scotia Formation, an ore bed with a maximum thickness of approximately 4 m; site W9 is in a thin bed a little below the Scotia Formation; site W6 is in two thin beds above the Scotia Formation.

The Wabana Group overlies the Bell Island Group with only a minor disconformity, showing no evidence

TABLE 1. Bell Island paleomagnetic directions* and paleomagnetic poles

Site	Stratigraphic location	Sample	NRM direction $D°$	NRM direction $I°$	Cleaned direction (to 450°C) $D°$	Cleaned direction (to 450°C) $I°$	Paleomagnetic classification
W1	Dominion Fm.	1	30.5	−5.2	31.0	−12.4	A
		2	24.7	7.1	24.4	−2.3	A
W2	Thin bed below Dominion Fm.	3	25.8	3.1	26.1	−12.7	A
		4	26.9	3.6	26.6	−10.0	A
W3	Scotia Fm.	5	10.8	1.0	13.2	−9.9	A
		6	14.4	5.9	15.2	−4.2	B
		7	18.2	0.9	19.6	−13.7	B
W4	Above Dominion Fm.	8	—	—	—	—	Unstable
W5	Dominion Fm.	9	12.5	8.8	18.7	−18.4	B
		10	16.5	18.8	24.1	−9.4	B
		19	19.2	−10.4	15.9	−26.3	A
W6	Thin bed above Scotia Fm.	11	39.1	29.9	32.0	2.5	C
		12	11.5	13.1	33.1	−9.9	C
		13	15.8	24.8	11.1	3.3	C
W7	Dominion Fm.	14	23.3	10.8	23.6	−4.8	A
		15	11.4	−3.9	14.7	−15.9	A
		16	26.9	5.7	26.0	−8.9	A
W8	Dominion Fm.	17	19.1	−4.6	17.8	−16.4	B
		18	12.1	−3.7	13.1	−20.9	B
W9	Thin bed below Scotia Fm.	20	53.0	19.2	40.5	−13.2	C
		21	22.1	46.2	11.2	10.2	C
Mean (unit weight to A, B samples)					20.7	−12.5	$k = 91$ $\alpha_{95} = 4.0°$
Corresponding paleopole					33°N	102°E	dm = 4.1° dp = 2.1°
Mean (unit weight to A, B sites)					21.2	−12.5	$k = 137$ $\alpha_{95} = 5.7°$
Corresponding paleopole					33°N	102°E	dm = 5.8° dp = 3.0°

NOTES: D and I are declination and inclination, respectively; k is Fisher's precision parameter; α_{95} is the radius of the circle of 95% confidence about the mean direction; and dm, dp are the semi-axes of the oval of 95% confidence about the paleopole.

*Directions with respect to bedding.

of any significant interval of non-deposition (Rose 1952). All beds sampled are of Early Ordovician age, probably early Arenigian (Bergstrom 1976).

All samples were oriented with the aid of a magnetic compass (the ores being sufficiently non-magnetic). The dip of each oolitic hematite bed sampled was measured; the beds dipped between 10 and 12° in a direction of azimuth between 310 and 340° (except for site W4, where the beds dipped 6° in a direction of azimuth 50°). All remanence directions quoted in this paper have been corrected for the dip of the beds, unless otherwise stated.

Laboratory procedure

In the laboratory, cylindrical specimens (1.9 cm in length and 2.3 cm in diameter) were cored from each block sample perpendicular to the bedding plane. One specimen from each sample was thermally demagnetized in stepwise fashion to peak temperatures of 200, 300, 350, 400, 450, 500, 550, 600, 640, 660 (or 665), and in most cases 680°C. After each cleaning step, specimen magnetizations were measured using a commercial spinner magnetometer.

Four to eight specimens were processed during a single experimental heating, using a furnace whose temperature could be maintained within ±3°C while heating at peak temperatures for 30 min. Thermal gradients over the volume occupied by eight specimens in the furnace were reduced by covering the specimens with a perforated copper pot when heating at or above 450°C. The variation of temperature between any two specimen locations in the furnace did not exceed 5°C. Three orthogonal Helmholtz coil pairs cancelled the Earth's magnetic field at the furnace. Because of extensive chemical alterations observed during heating of some samples (see below), field cancellation was continuously monitored to ensure as close to zero residual field as possible. Cancellation over the volume occupied by eight specimens was effective to within 25γ during the experimental run.

Based on the experimental results of the paleomagnetic work, representative samples were selected to study (a) the degree of chemical alterations as a function of temperature using Curie balance experiments, and (b) possible anisotropy of acquisition of thermoremanent magnetization (TRM).

In the TRM experiment, a previously untreated specimen was thermally demagnetized at one or more temperatures. These temperatures were selected from Curie point data to avoid exceeding the temperature at which the onset of chemical alterations becomes apparent. After each demagnetization step, the specimen was again heated to the same peak temperature, but cooled in the Earth's magnetic field. The direction of the partial TRM (pTRM) acquired during the second step was then compared with that of the Earth's magnetic field.

In the Curie balance experiments, whole-rock chips were heated in air in a magnetic field of 1200 Oe (95 $kA \cdot m^{-1}$). Magnetization was measured as a function of temperature during heating and cooling. Temperature was cycled to higher and higher maximum temperatures to identify the temperature intervals during which most chemical alterations occur and the Curie points of the magnetic minerals produced.

Experimental results

Thermal demagnetization studies

Experimental results of the thermal demagnetization studies are summarized in Table 1. The single sample collected at site W4 shows no directional stability and, hence, is not considered further. At the remaining eight sites NRM directions initially have shallow inclinations and north-northeast declinations. Upon stepwise thermal cleaning the directions move systematically to steeper negative inclinations, often reaching stable end points in peak temperatures above 400°C. However, a large percentage of the samples acquire large randomly oriented components as a result of the production of "magnetite" in the higher temperature steps (see below). Although in a few samples these changes do not occur in the range of temperature utilized, in others the severe chemical alterations occur at temperatures as low as 450°C. Rao (1970) also observed large increases in intensity, but no corresponding randomization of directions.

The behaviour of individual samples can be classified on the basis of their paleomagnetic usefulness and the temperature at which large-scale chemical alteration effects are observed (see summary in Table 1).

"A" samples are those that maintain a consistent directional pattern (a stable end point and (or) a systematic directional "swing") to peak temperature $\geq$ 600°C. Nine samples fall within this category. Examples are illustrated in Fig. 2. Invariably directions of A samples move from shallow to somewhat steeper negative inclinations before randomizing in the 660 (665) or 680°C step. Intensities fall smoothly to 10 or 20% of the initial NRM intensity at the highest temperatures. Hematite is clearly indicated as the dominant remanence carrier.

Six "B" samples maintain consistent directional patterns to peak temperatures of 500–600°C before components caused by chemical alterations begin to mask the NRM (see Fig. 3). The chemical alterations are accompanied by random directions, and intensity increases often several times that of the NRM. Fortunately, a stable end point (similar to that observed for A samples) is usually attained at about 400°C, so that B samples are of use in establishing a paleodirection.

Five "C" samples (see Fig. 4) are of lower stability, having large "viscous" components that are not fully erased in the stepwise thermal cleaning process before the NRM is obscured by the large random components produced by chemical alteration. Magnetization directions of C samples swing along great-circle paths toward the end points established in A and B samples, but the swings are incomplete. Hence, these samples are considered unsuitable for paleomagnetic statistical purposes.

pTRM acquisition experiments

Four samples (1, 3, 5, and 15) representative of the A-type behaviour that is observed in thermal demagnetization experiments and single samples representing each of the B- and C-types were given pTRM's in the present Earth's magnetic field in order to study their ability to acquire remanence along the direction of an applied field. Peak temperatures were selected to allow maximum remanence acquisition and yet avoid the effects of the chemical alterations described below.

The A samples were given pTRM's in cooling from peak temperatures of 300, 450, and 600°C, since most of the chemical alteration observed in Curie-point experiments occurs above 600°C. The B sample was given pTRM's from peak temperatures of 300 and 400°C, and the C sample from 300°C only.

Results are illustrated in Fig. 5. At 300°C the directions of the pTRM show a small degree of scatter, undoubtedly due to the low intensity of the magnetization acquired in these low temperatures (23% of NRM intensity). Neverthless the mean direction ot the pTRM acquired in the six samples does not differ significantly at the 95% confidence level from the direction of the applied field. Similarly, at 450°C, where intensities are significantly enhanced (82% of NRM intensity) and directional scatter correspondingly decreased, the pTRM is closely aligned with the direction of the applied field. The results at 600°C are not considered, since anomalously high intensities (47 times the NRM inten-

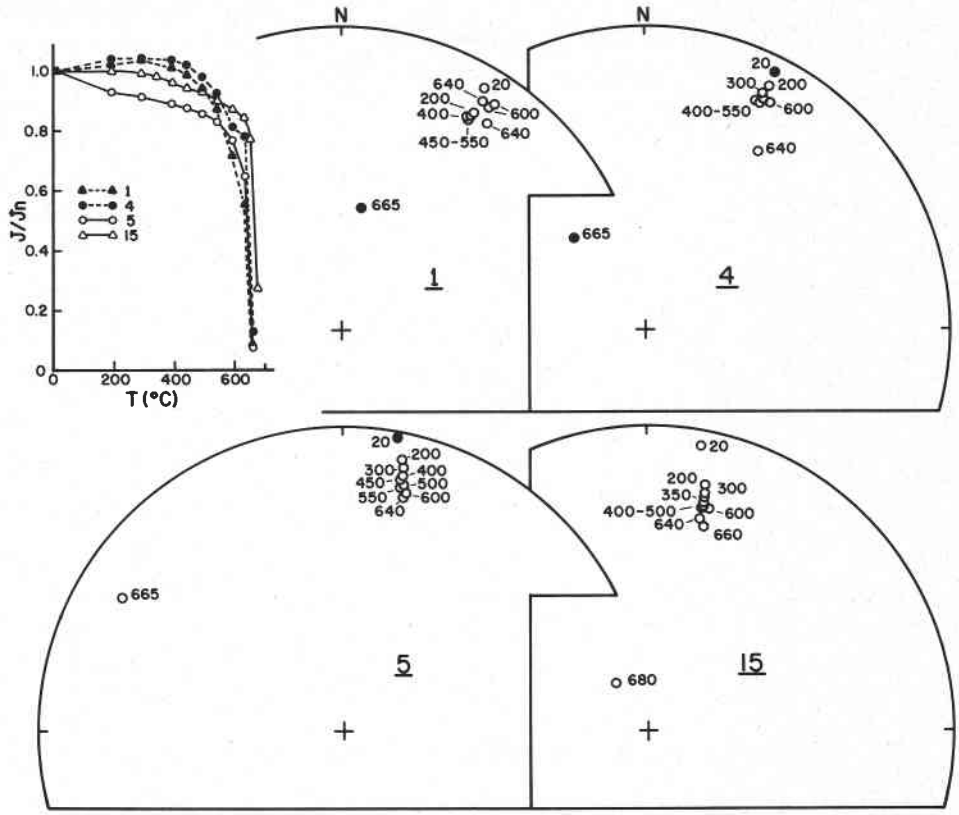


FIG. 2. Characteristic directional and intensity results for A-type samples. Directions are plotted on a stereographic (equal-angle) projection with open (closed) symbols indicating negative (positive) inclinations.

sity) indicated the occurrence of significant chemical alterations.

Curie-balance experiments

Thermomagnetic curves were measured for the three C samples of Fig. 4 and the four A samples of Fig. 2. The paleomagnetically unsuitable C samples possess distinctly different thermomagnetic curves from the paleomagnetically acceptable A samples, as shown for one typical C sample and one typical A sample in Fig. 6.

The three C samples all show that a strongly magnetic mineral with a Curie temperature of ~610°C is being produced on heating to 700°C. This magnetic mineral raises the sample's intensity of magnetization in 1200 Oe (95 $kA \cdot m^{-1}$) by ~18–~90 times. At least a quarter of this magnetic mineral has been produced on heating to 450°C, and at least another third is added between 450 and 500°C. Thus one might expect thermal demagnetization even at 450°C to be very difficult for C samples.

In contrast, for the four A samples measured, heating to 600°C and cooling increase the magnetization in 1200 Oe (95 $kA \cdot m^{-1}$) by only about one to three times. A larger increase in magnetization in 1200 Oe is added by heating from 600 to 700°C and cooling; in those A samples where this increase is largest, it is mainly due to the growth of a strongly magnetic mineral with a Curie point of ~580°C, presumably magnetite. Thus one would expect that A samples would be relatively easy to thermally demagnetize to 600°C, but that above 600°C some A samples might acquire spurious components of magnetization.

Interpretation of Curie-balance experiments

Upon heating, the three C samples measured all show growth of a strongly magnetic mineral with a Curie point of ~610°C, a little higher than expected of stoichiometric magnetite (and thought to be due to maghemite by Rao (1970)). Almost all of this growth of "magnetite" occurs between 300 and 600°C, with the interval 450–500°C being especially productive. Through thin-section examination and chemical analysis of the oolitic hematite ores, Hayes (1915) showed that chamosite (15–25%) and calcium phosphate of brachiopod shell fragments (4–5%) usually accompanied the dominant hematite (50–70%). Other more variable constituents

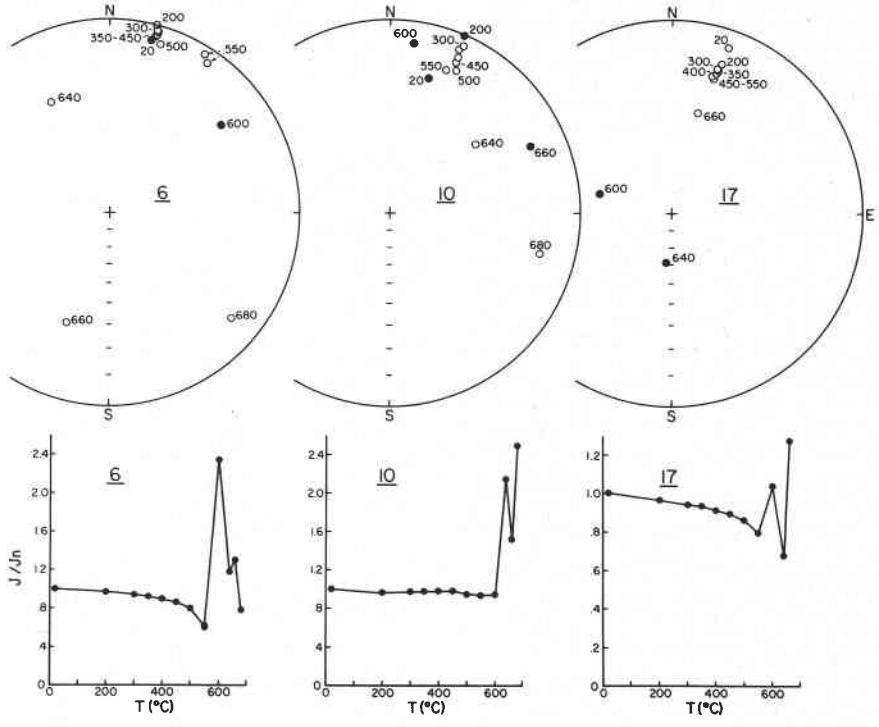


FIG. 3. Characteristic directional and intensity results for B-type samples. Symbols as in Fig. 2.

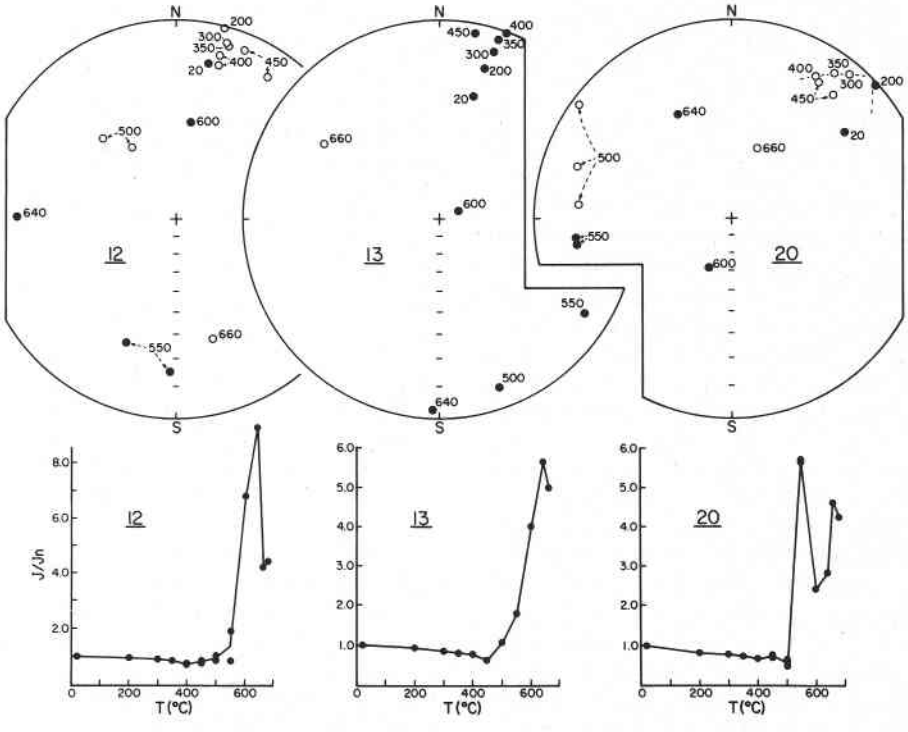


FIG. 4. Characteristic directional and intensity results for C-type samples. Symbols as in Fig. 2.

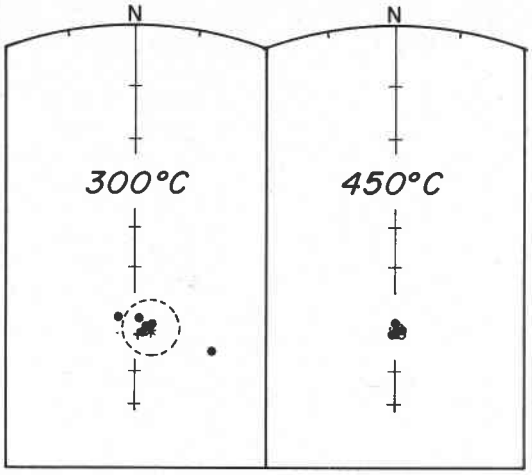


FIG. 5. Directions of pTRM's acquired in the Earth's magnetic field (shown as a cross). All directions are downward.

included siderite (0–50%), quartz (0–10%), calcite (0–1%), and minor pyrite. (All percentages quoted in this section of the paper are percentages by weight.) This mineralogy is supported by X-ray diffractograms of the six samples of Table 2. With this mineralogy, four possible mechanisms for production of "magnetite" upon heating deserve consideration: (1) production by thermal decomposition of siderite (the mechanism suggested by Rao 1970); (2) production by high-temperature oxidation of pyrite; (3) production by thermal decomposition of chamosite (French and Van der Voo 1979; Barrie *et al.*1980); and (4) production by action upon hematite of hot water vapour from decomposition of chamosite.

Heating pyrite in air in our Curie balance did produce a mineral with a Curie point of ~600°C, though most of this production occurred above 500°C. Judging from polished sections of the six samples listed in Table 2, the C samples on average seem to contain more pyrite than the A samples. However, none of the six samples contains more than 1% pyrite, and individual A and C samples (samples 1 and 13) seem to contain roughly the same amount of pyrite. Hence, pyrite is unlikely to account for the very large growth of "magnetite" in C samples below 500°C.

Chamosite (ferrous or ferric) heated in air is reported to become amorphous at ~450°C, with hematite beginning to crystallize out at ~550°C (Brindley and Youell 1953). Though one might expect reducing conditions to produce magnetite in place of hematite, the experiments of Brindley and Youell suggest not. However, French and Van der Voo (1979) and Barrie *et al.* (1980) reported that chamosite in Rose Hill sandstone does decompose to magnetite (among others) on heating to

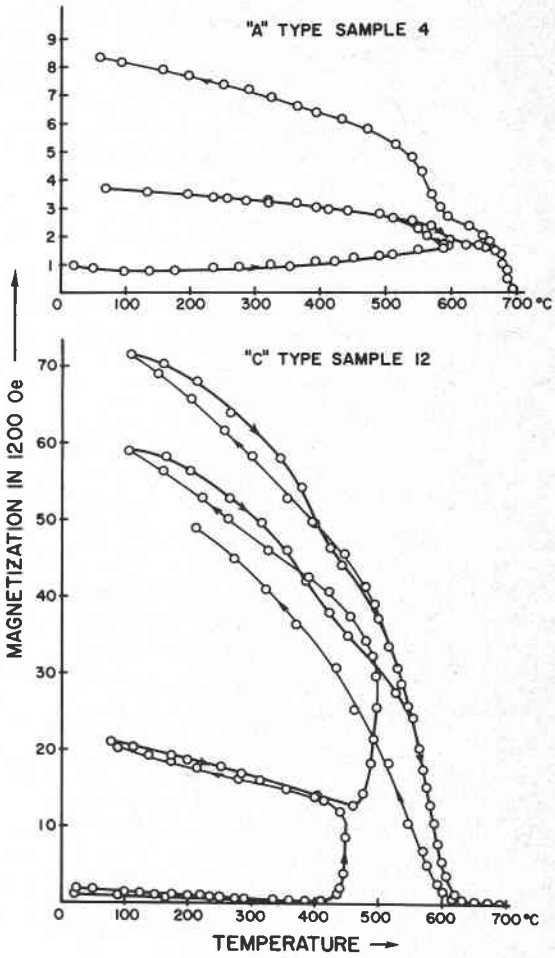


FIG. 6. Contrasting the thermomagnetic curve in 1200 Oe (95 $kA \cdot m^{-1}$) of a typical A sample (heated successively to 600 and 700°C) with that of a typical C sample (heated successively to 400, 450, 500, 550, and 700°C; the heating intervals 550–600, 600–650, and 650–700°C are not illustrated, but each caused net reductions in magnetization). Heating lines are shown fuller than cooling lines. All curves are normalized to 20°C before heating.

600°C. There are similar amounts of chamosite in both A and C samples judging from similar Al_2O_3 contents of A and C samples (Table 2) and from the similar heights of peaks due to chamosite in our X-ray diffractograms of A and C samples (using hematite peaks for normalization). Almost all of the chamosite in both A and C samples has decomposed by 500°C, judging from X-ray diffractograms. Hence, if chamosite does decompose to magnetite, it should produce a similar amount in both the A and C samples, making this an unlikely mechanism for the much larger production of "magnetite" in C samples.

Chamosite (ferrous) is reported to lose most of its

TABLE 2. Chemical analyses of Wabana – Bell Island oolitic hematite samples

Sample	Weight percent of oxide in sample									Total	Weight percent carbon in sample
	SiO_2	TiO_2	Al_2O_3	Fe_2O_3*	MnO	MgO	CaO	P_2O_5	LOI†		
1	34.9	0.37	3.46	57.43	0.03	0.41	1.09	0.94	1.85	100.48	0.06
4	23.6	0.30	3.85	66.73	0.02	0.48	1.35	1.02	1.77	99.12	0.10
5	11.6	0.19	3.44	75.88	0.02	0.41	1.93	1.71	3.53	98.71	0.14
12	10.1	0.26	6.92	67.84	0.34	0.84	2.64	1.70	7.92	98.56	1.65
13	8.87	0.24	6.07	67.34	0.40	0.86	3.24	1.68	10.41	99.11	2.22
20	11.8	0.32	5.82	65.33	0.30	1.25	3.58	1.95	8.55	98.90	2.11

*Total iron is given as Fe_2O_3.
†LOI gives loss on ignition.

structural water at ~400°C (Brindley and Youell 1953). This hot water vapour might react with hematite to produce "magnetite," since Johnson *et al.* (1972) point out that a magnetite-like spinel with a Curie point of ~595°C is produced on heating wet hematite powder in air. However, we have shown above that both A and C samples contain a similar amount of chamosite that decomposes by 500°C. The chamosite content of A and C samples (estimated assuming all of the Al_2O_3 of Table 2 is in chamosite) should liberate ~2% by weight of water in the samples on decomposing. Both A and C samples do appear to lose ~2% by weight of water on ignition (correcting the loss on ignition measurements of Table 2 for CO_2 loss, assuming all of the carbon in the samples is converted to CO_2). Hence, the hot water vapour should convert similar amounts of hematite into "magnetite" in both A and C samples, making this an unlikely mechanism for the much larger production of "magnetite" in C samples.

Siderite heated in air decomposes to a very magnetic black mineral, which according to Rowland and Jonas (1949) is a spinel with a diffraction pattern similar to that of magnetite. Unlike pyrite or chamosite, siderite is very abundant in all three of the C samples of Table 2 and almost absent in the three A samples of Table 2, making siderite decomposition a likely mechanism for the much larger "magnetite" production in C samples. The average siderite content of the three C samples is estimated as ~17% by weight, whereas that of the three A samples is estimated as less than 1% using the chemical analyses of Table 2 (assuming that all of the carbon is in siderite and calcite, all of the CaO in calcite and calcium phosphate, and all of the P_2O_4 in calcium phosphate). The much larger siderite content of C samples is also supported by the X-ray diffractograms, whose siderite peaks are an order of magnitude higher for the three C samples than for the three A samples (hematite peaks providing normalization). The siderite peaks barely rise above the noise level for the A samples.

Though it is probable that most of the "magnetite" produced on heating C samples is due to decomposition of siderite, there are two difficulties with this mechanism. One difficulty is that differential thermal analysis by Rowland and Jonas (1949) shows that siderite decomposes at ~580°C, whereas most of the "magnetite" in our C samples appears below 500°C. The difference may be due to water from dehydration of chamosite. Whereas dry powdered siderite (from Ivigtut, Greenland) that we heated in air in our Curie balance produced almost all of its "magnetite" above 500°C, the same powdered siderite moistened with water produced almost all of its "magnetite" below 500°C. The other difficulty is that the Curie point of the "magnetite" produced from the Ivigtut siderite was much lower than the ~610°C Curie point of the "magnetite" produced in C samples. The Ivigtut siderite heated in air produced "magnetite" of ~555°C Curie point (~535°C on cooling) when dry and ~525°C Curie point (~490°C on cooling) when wet. The higher Curie point of the "magnetite" produced in C samples may be due to the siderite being in the presence of hematite. The powdered Ivigtut siderite mixed with synthetic hematite powder and heated dry in air produced "magnetite" of ~595°C Curie point (most of the "magnetite" transforming to a ~515°C Curie point on cooling).

Since siderite is likely responsible for the difficulty of thermally demagnetizing C samples, leaching in HCl before thermal demagnetization may improve C samples by removing siderite more quickly than hematite. The growth of "magnetite" in a chip of sample 12 did seem to be greatly reduced by leaching the chip in dilute HCl before heating.

Statistical analysis

All C samples were excluded from the statistical analysis of the paleomagnetic directional data on account of the difficulties experienced in isolating a stable end point (see above). At each step in the thermal

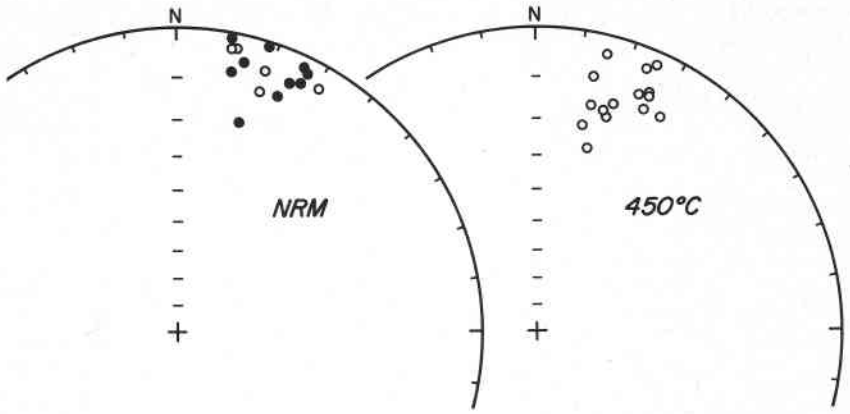


FIG. 7. Directions of magnetization of each of the 15 Wabana – Bell Island samples used for statistical purposes.

demagnetization procedure, a mean direction of magnetization and a circle of 95% confidence were calculated for the 15 A and B samples using standard Fisher statistical methods.

The radius of the circle of 95% confidence, α_{95}, decreases slowly from a value of 4.7° at 20°C to 4.0° at 300 and 450°C, then slowly increases to 5.3° at 550°C. At 600°C, α_{95} jumps to 21.4°, consistent with the acquisition of large spurious components in several samples. Similarly, the precision parameter k attains its highest values, 90.5 and 90.6, at 300 and 450°C, respectively. As some samples have yet to reach a stable end point at 300°C, the 450°C step was selected as that most representative of the paleofield direction. Directions of magnetization of each of the 15 samples used for statistical purposes are shown in Fig. 7.

Unit weight to each of the 15 A and B samples yields a mean direction of $D = 20.7°$, $I = -12.5°$ ($k = 91$, $\alpha_{95} = 4.0°$). However, a site mean direction was also calculated, since the radius of the circle of confidence is artificially lowered by taking sample rather than site means. The site mean direction $D = 21.2°$, $I = -12.5°$ differs only slightly from the sample mean, but the corresponding α_{95} (5.7°) is thought to be more representative of the reliability and scatter of the data. A paleopole corresponding to the Wabana Group and Bell Island Group site mean direction lies at 33°N, 102°E (dm = 5.8°, dp = 3.0°).

It should be noted that, although the mean direction obtained after thermal demagnetization to 450°C has been used to calculate the paleopole for the Wabana – Bell Island rocks, the mean direction obtained at higher temperatures is not significantly different. For example, at 550°C, unit weight to each of the 15 A and B samples gives a mean direction at $D = 23.3°$, $I = -13.7°$ ($k = 53$, $\alpha_{95} = 5.3°$). At 640°C, B samples are no longer stable, but unit weight to each of the nine A samples gives a mean direction of $D = 30.0°$, $I = -19.3°$ ($k = 26$, $\alpha_{95} = 10.3°$). In the latter case, intensities of magnetization have fallen to an average of 75% of their NRM value, with no indication of the presence of any other higher blocking temperature component. By 660°C, the A samples show a large scatter in directions ($D = 9.8°$, $I = -8.2°$, $k = 3.3$, $\alpha_{95} = 33.7°$), with their average intensity reduced to 43% of NRM (though for samples 1, 4, and 5, intensities have dropped to ~10% of NRM).

Interpretation

Early Paleozoic paleopoles for North America are generally more poorly grouped than those obtained from younger rocks (e.g., Irving 1979). A number of explanations have been suggested for the greater scatter of the older data. Kent and Opdyke (1978), Van der Voo *et al.* (1979), and Irving (1979) attribute the scatter of Devonian and Early Carboniferous poles to a difference between those from cratonic North America and those from the Atlantic region of New England and the Maritimes, and postulate that the coastal region was located well south of its present position relative to cratonic North America until Late Carboniferous times. Watts *et al.* (1980) interpret the large scatter of Cambrian poles obtained in the stable craton as representing an extensive loop in the North American polar wander path during Cambrian times.

For the Ordovician Period, paleomagnetic coverage is, as yet, incomplete. Poles obtained from Ordovician rock units come, for the most part, from cratonic North America, including the western part of Newfoundland. With the exceptions of the earlier poles for the Wabana Group (Rao 1970; Deutsch and Rao 1977), and a pole for the Cambrian Arden pluton of Delaware, which is believed to have been remagnetized during the Ordovician (Rao and Van der Voo 1980), no results of this age are available from the Atlantic region or the Avalon zone of eastern Newfoundland. Hence the Wabana – Bell

TABLE 3. Ordovician, Silurian, and Devonian paleopoles for cratonic and Atlantic North America

	Geological unit	Age	Paleopole (°)	Reference
1a.	Front Range intrusions, Colorado (AF)	Cambr.–Ord.	37N 122E	Larson and Mutschler (1971)
1b.	(t)		44N 100E	Larson and Mutschler (1971)
2.	Trenton limestone, New York	Ord.	36N 114E	McElhinny and Opdyke (1973)
3.	Beemerville Complex, New Jersey	Ord.	35N 126E	Proko and Hargraves (1973)
4.	St. George dolostone and sphalerite ore, Newfoundland	Ord.	26N 126E	Beales *et al.* (1974)
5.	Steel Mountain anorthosite, Newfoundland	Ord.	22N 139E	Murthy and Rao (1976)
6.	St. George limestone, Newfoundland (tilt-corrected)	Ord.	22N 120E	Deutsch and Rao (1977)
7.	Moreton's Harbour basalt, Newfoundland	Ord.	32N 130E	Deutsch and Rao (1977)
8.	Juniata Formation, Pennsylvania, West Virginia and Virginia	Ord.	32N 114E	Van der Voo and French (1977)
9.	Moccasin–Bays Formations, Tennessee	Ord.	33N 147E	Watts and Van der Voo (1979)
10.	Chapman Ridge Formation, Tennessee	Ord.	27N 112E	Watts and Van der Voo (1979)
11.	Arden pluton, Delaware	Ord.	16S 123E	Rao and Van der Voo (1980)
12.	Weedon–Gould area rocks, Quebec	Ord.	17N 164E	Seguin (1980*a*)
13.	Bloomsburg Formation, Pennsylvania	Sil.	32N 102E	Roy *et al.* (1967)
14.	Mascarene Group, New Brunswick	Sil.	02S 122E	Roy *et al.* (1979)
15.	Botwood Group rhyolite, Newfoundland	Sil.	13N 125E	Lapointe (1979)
16.	Botwood Group sandstone, Newfoundland	Sil.	25N 113E	Lapointe (1979)
17.	Mt. Peyton diorite, Newfoundland	Sil.	15S 068E	Lapointe (1979)
18.	Rose Hill Formation, Maryland, West Virginia, and Virginia	Sil.	19N 129E	French and Van der Voo (1979)
19.	Weedoon–Gould metavolcanics, Quebec	Sil.	29N 125E	Seguin (1980*a*)
20.	Cape Breton granites, Nova Scotia	Sil.	16S 139E	Rao *et al.* (1981)
21.	St. Francis metasedimentary group, Quebec	Sil.–Dev.	24N 149E	Seguin (1980*a*)
22.	Hersey Formation, Maine	Sil.–Dev.	20N 129E	Kent and Opdyke (1980)
23.	Scotstown intrusive, Quebec	Sil.–Dev.	40N 144E	Seguin (1980*b*)
24.	Cape Breton granites, Nova Scotia	Sil.–Dev.	32S 083E	Rao *et al.* (1981)
25.	Perry Formation red beds, New Brunswick	Dev.	32N 118E	Robertson *et al.* (1968)
26.	Raisin R. dolomite and Columbus limestone, Ohio	Dev.	24N 164E	Martin (1975)
27.	Columbus limestone, Ohio	Dev.	45N 120E	Martin (1975)
28.	Delaware limestone, Ohio	Dev.	48N 118E	Martin (1975)
29.	Mafic complex, Massachusetts	Dev.	23N 126E	Schutts *et al.* (1976)
30.	Temple Butte limestone, Arizona	Dev.	53N 115E	Elston and Bressler (1977)
31.	Beckers Butte and Jerome Members, Arizona	Dev.	56N 109E	Elston and Bressler (1977)
32.	Catskill Formation, New York	Dev.	47N 117E	Kent and Opdyke (1978)
33.	Mt. Peyton granite, Newfoundland	Dev.	63S 125E	Lapointe (1979)
34.	St. George pluton, New Brunswick	Dev.	38S 085E	Roy *et al.* (1979)
35.	St. George pluton, New Brunswick	Dev.	29N 093E	Roy *et al.* (1979)
36.	St. Stephen pluton, New Brunswick	Dev.	43S 136E	Roy *et al.* (1979)
37.	Catskill Formation, Pennsylvania, Maryland, and W. Virginia	Dev.	44N 124E	Van der Voo *et al.* (1979)
38.	Dockendorff Group, Maine	Dev.	24N 084E	Brown and Kelly (1980)
39.	Eastport Formation	Dev.	24N 114E	Kent and Opdyke (1980)
40.	Cape Breton granites, Nova Scotia	Dev.	27N 084E	Rao *et al.* (1981)
41.	Scotstown intrusive, Quebec	Dev.	34N 149E	Seguin (1980*b*)
42.	Scotstown intrusive, Quebec	Dev.	39N 131E	Seguin (1980*b*)

Island Group paleopole, if representative of the Ordovician paleofield, is critical for establishing the relative locations of Atlantic North America and cratonic North America at that time.

The age of the Wabana – Bell Island magnetization, however, is not certain. The remanent magnetization of our oolitic hematite specimens is probably a chemical remanence acquired when the hematites formed. Yet, oolitic hematites in many Precambrian and Paleozoic deposits may have originally been goethite that has been diagenetically transformed (Blatt *et al.* 1980). Ranger (1979) proposed that the Wabana – Bell Island oolitic hematites formed in the following manner. The oolites probably began as chamosite and have survived as such in rocks of the lagoonal facies. Chamosite oolites on tidal flats largely oxidized, presumably first producing goethite, which was transformed to hematite on diagenesis.

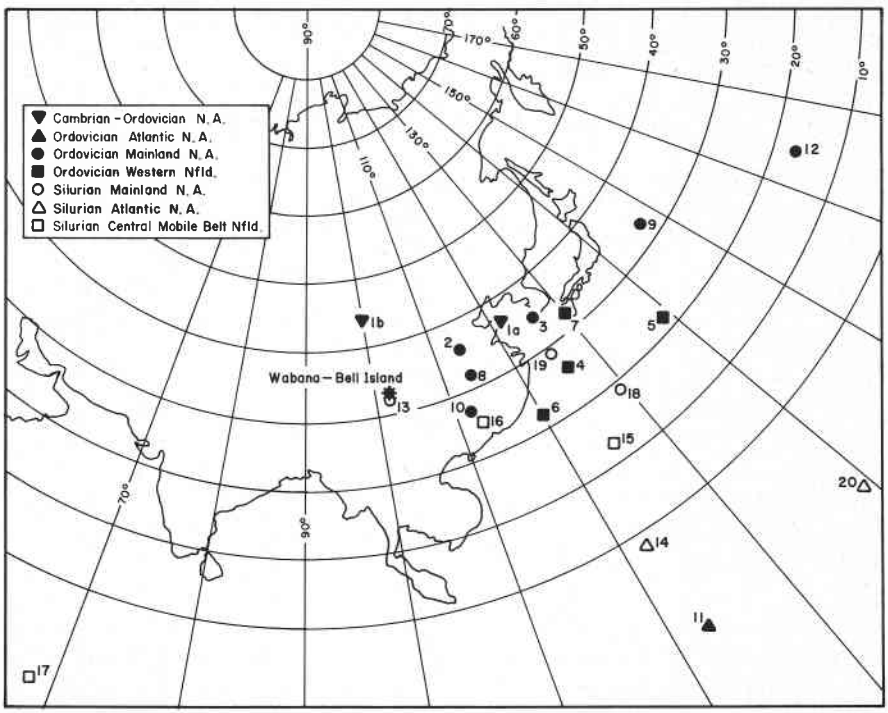


FIG. 8. Comparison of the Wabana – Bell Island paleopole with Ordovician and Silurian paleopoles for North America. Pole numbering as in Table 3.

Although the transformation to hematite probably occurred in the Early Ordovician, a later transformation is also possible. For example, goethite remains the main oxide in the Jurassic oolitic iron ores of Europe (Blatt *et al.* 1980). Transformation to hematite may have occurred as late as the low-grade Devonian metamorphism that affected the Avalon Peninsula. (On Bell Island, the metamorphism is very likely of lower grade than the prehnite–pumpellyite facies found by Papezik (1969) in nearby more deformed Hadrynian ignimbrites.) Unfortunately, the rocks of Bell Island are too uniformly tilted for a fold test that would determine whether the remanence was acquired before or after the probable Devonian deformation.

The timing of acquisition of remanence is critical in interpreting the Wabana – Bell Island result paleogeographically. Hence, we will consider in turn possible Ordovician and Devonian ages for our Wabana – Bell Island magnetization.

North American paleopoles of Ordovician, Silurian, and Devonian ages are listed in Table 3 and plotted in Figs. 8 and 10. Several published poles are not considered. These include: the pole for the Silurian Clinton Formation of Alabama (Howell *et al.* 1958), for which no demagnetization was carried out; results from Ordovician volcanic rocks of Quebec (Seguin 1977), which are difficult to interpret because of structural complexity and because AF demagnetization yields a paleopole widely divergent from that obtained on thermal demagnetization; a pole based on only two Ordovician dikes in the Grenville Province of Ontario (Palmer *et al.* 1977); and a result from the Devonian Belchertown intrusion of Massachusetts, which is suspected of having undergone significant rotation (Ashwal and Hargraves 1977). In addition, results published only in abstract form are omitted.

Most Ordovician paleopoles form a scattered grouping centred near 30°N 120°E (Fig. 8). Exceptions include pole 12 (Weedon–Gould area of Quebec), which falls well east of the main grouping, and pole 11 (Arden pluton of Delaware), which lies ~40° south and may indicate (Rao and Van der Voo 1980) that the Arden area was not part of North America at the time of its magnetization. Likewise, Silurian paleopoles from cratonic North America and central Newfoundland fall in or close to the Ordovician grouping (except pole 17 from the Mount Peyton diorite of Newfoundland). Two Silurian poles obtained from the Avalon zone, pole 14 (Mascarene Group, New Brunswick) and pole 20 (Cape Breton granites, Nova Scotia), lie well south.

Ordovician poles from western Newfoundland are generally a little south and east of those from mainland Canada and the U.S.A. The pole for the Wabana – Bell Island Groups of eastern Newfoundland (present study),

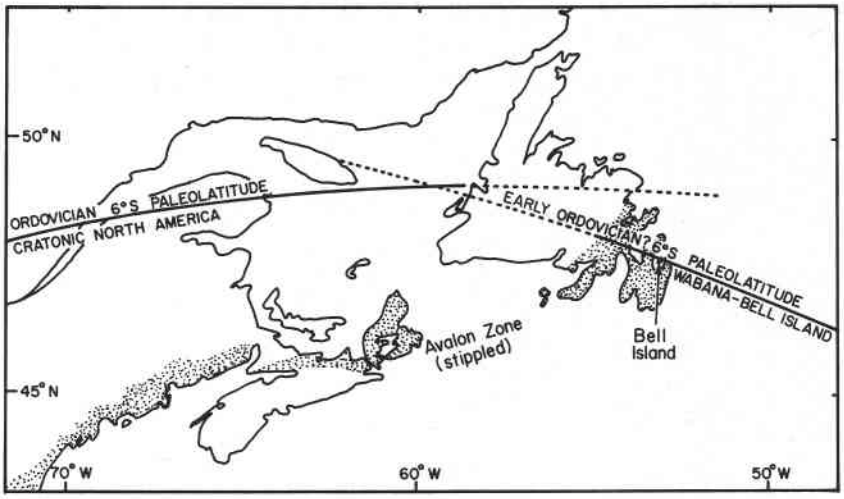


FIG. 9. Comparison of the 6°S Early Ordovician (?) paleolatitude line from the Wabana – Bell Island paleopole with the 6°S Ordovician paleolatitude line for mainland North American paleopoles.

however, falls just west of the main grouping of Ordovician mainland poles (Fig. 8). It is almost identical with the pole of the Silurian Bloomsburg Formation of Pennsylvania (pole 13).

Assuming an Early Ordovician age for the Wabana – Bell Island magnetization, the paleolatitude of Bell Island (and the Avalon zone) can be computed and compared with that for cratonic North America. Figure 9 shows the 6°S (±3°) Early Ordovician paleolatitude for the Avalon zone based on the Wabana – Bell Island paleopole. Figure 9 also shows the 6°S (±7°) Ordovician paleolatitude line for cratonic North America, based on poles from the main grouping of Ordovician mainland poles (poles 2, 3, 8, 9, and 10 of Table 3). It was determined in the following manner. The direction of magnetization corresponding to each of these mainland poles was calculated for a single geographical location and a Fisherian mean direction was obtained. The corresponding paleopole (34°N 123°E) was then calculated and its 6°S paleolatitude line was sketched.

As can be seen from Fig. 9, the paleomagnetic evidence, based on an Early Ordovician age for the Wabana – Bell Island pole, suggests that the Avalon zone occupied a similar Early Ordovocian paleolatitude as the presently adjoining North American mainland. It also had a similar orientation as at present relative to the mainland (although a small clockwise rotation by 17° of the sampling site relative to the mainland since the Early Ordovician is possible).

Paleomagnetic evidence, however, does not rule out the possibility that an Iapetus Ocean of undetermined width existed between the Avalonian microcontinent and the North American continent during the Early Ordovician and that the ocean had closed by the Silurian as suggested by geological evidence (Williams 1979). Assuming that the Iapetus Ocean between North American and the Avalonian microcontinent was wide in the Early Ordovician and that it had closed by the Silurian, the paleomagnetic evidence would require that this part of the Iapetus Ocean closed approximately along the Ordovician paleolatitude lines.

Kent and Opdyke (1978), however, have suggested, on the basis of paleomagnetic evidence, that the Avalonian microcontinent was located ~20° south of its present location relative to cratonic North America during the Devonian. If so, an Early Ordovician age for the Wabana – Bell Island pole would require the Avalonian microcontinent to have moved ~20° south relative to cratonic North America between the Early Ordovician and the Devonian—perhaps during the closing of the Iapetus Ocean.

As noted above, the Wabana – Bell Island rocks may have acquired their remanence long after formation, perhaps during Devonian metamorphism. Hence, comparison of the Wabana – Bell Island paleopole with Devonian poles from the Avalon zone and with those from cratonic North America is in order (though the Wabana – Bell Island remanence might then be post-folding, changing the paleopole a little to 36.6°N, 102°E).

Devonian and Siluro-Devonian paleopoles from North America are plotted in Fig. 10. Most poles from the craton fall on a northwest–southeast band between about 55°N 110°E and 25°N 165°E. However, poles from the Avalon zone of Atlantic North America lie, without exception, to the south of this band.

A number of Avalon zone poles (poles 22, 25, 29, 35, 39 and 40), including that for the Wabana – Bell Island

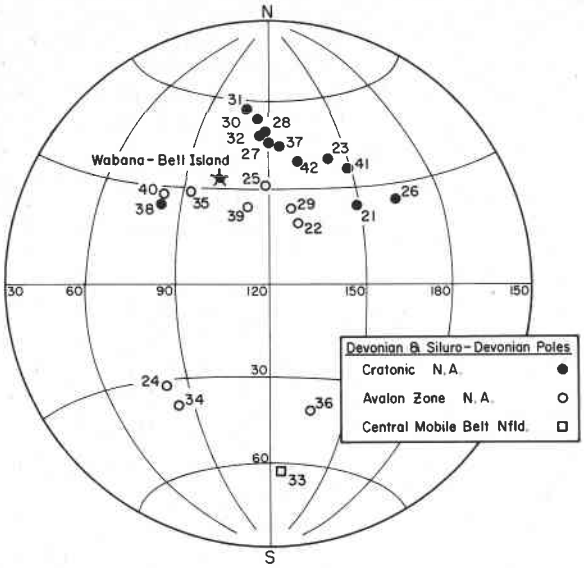


FIG. 10. Comparison of the Wabana – Bell Island paleopole with Silurian–Devonian and Devonian paleopoles for North America. Pole numbering as in Table 3.

Groups, are scattered perhaps 20° south of the cratonic polar grouping. This difference and a similar discrepancy between Early Carboniferous poles from the craton and the Avalon zone led Kent and Opdyke (1978) and Van der Voo *et al.* (1979) to postulate a displacement of the Avalon zone about 20° south of its present location relative to the North American craton during Devonian and Early Carboniferous times. More recent data, however, have not always conformed to this rather simple pattern. For example, poles 24, 34, and 36 reported from the Avalon zone and pole 33 from the Central Mobile Belt of Newfoundland fall ~60° south of the main grouping of Avalon zone poles.

In recent reviews (Irving 1979; Roy *et al.* 1979; Kent and Opdyke 1980; and Rao *et al.* 1981), possible explanations of the widely scattered Devonian poles from the Avalon zone have been considered. These include a displaced Avalonian terrain, rapid polar wander, post-magnetization tilting of the intrusions responsible for the most divergent poles, insufficient resolution of multicomponent magnetizations or a combination of the above mechanism. All authors, however, conclude that more data for both the Avalon zone and cratonic North America will be necessary before a satisfactory explanation of the scatter of Devonian poles is achieved.

Conclusion

The age of the paleopole calculated for the Lower Ordovician Wabana and Bell Island Groups (33°N 102°E) is uncertain. If of Early Ordovician age, the Wabana – Bell Island result indicates that the Avalon zone of eastern Newfoundland occupied a similar Early Ordovician paleolatitude as the presently adjoining North American mainland. If of Devonian age, it is not inconsistent with many other Devonian poles for the Avalon zone of Atlantic North America. The difference between these poles and Devonian poles from the North American craton may indicate that the Avalon zone was displaced ~20° south of its present position relative to the craton in Devonian times. However, such a conclusion may be premature in light of the large scatter of more recently reported Devonian poles from the Avalon zone.

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